



Evaluating 2D and 3D-Aware Vision Foundation Models for Vehicle Attribute Recognition

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Motivation

- Automatic License Plate Recognition (ALPR) performance degrades under **challenging real-world conditions**.
- Vehicle Attribute Recognition can provide a fallback to improve the robustness of vehicle recognition systems.
- We evaluate 2D and 3D-aware vision foundation models for Vehicle Attribute Recognition in a Brazilian surveillance scenario.



UFPR-VeSV dataset

- 24,945 images from the Military Police of Paraná (Brazil) surveillance system.
- Vehicles distributed across 14 types, 26 makes, and 136 models, presenting a severe long-tailed class distribution.
- Representative of challenging real-world conditions.

*Dataset developed in an independent study. Publicly available at <https://github.com/Lima001/UFPR-VeSV-Dataset>



Figure: Examples of adverse surveillance capture conditions present in the UFPR-VeSV dataset.





Benchmarked Models

Paradigm	Year	Model	Pre-training Objective
Standard 2D	2021	DINOv1	Self-distillation
	2022	iBOT(ImageNet-22K)	Masked Modeling
	2023	DINOv2	Self-distillation
	2024	DINOv3	Self-distillation
	2024	SAM 3	Segmentation
3D-Aware	2023	CroCo v2	Cross-view Completion
	2024	Depth Anything v1	Monocular Depth
	2024	Depth Anything v2	Monocular Depth
	2024	DUSt3R	Point Cloud Reg.
	2024	Gamba	3D Reconstruction
	2024	Hunyuan3D 2.1	3D Reconstruction
	2024	MASt3R	Stereo Matching
	2024	SAM 3D	3D Reconstruction
	2025	Depth Anything 3	Monocular Depth





A few questions

- Do these models generalize to our data?
 - Are their representations robust to real-world conditions?
 - Do 2D and 3D representations behave differently?
- Real-world imbalance → Macro-Accuracy (Ma-Acc)
 - Few-shot / low-data & OOD performance
 - Pose & IR sensitivity



Protocol

- Frozen feature extraction & linear probing.
- CLS token + GAP features when possible; for encoder-decoder architectures we utilize the encoder's output, otherwise the last latent vector.
- Type, Make, and Model classification are evaluated independently.
- Adopted metrics: Mi-Acc vs Ma-Acc, averaged over 10 runs across different dataset splits.

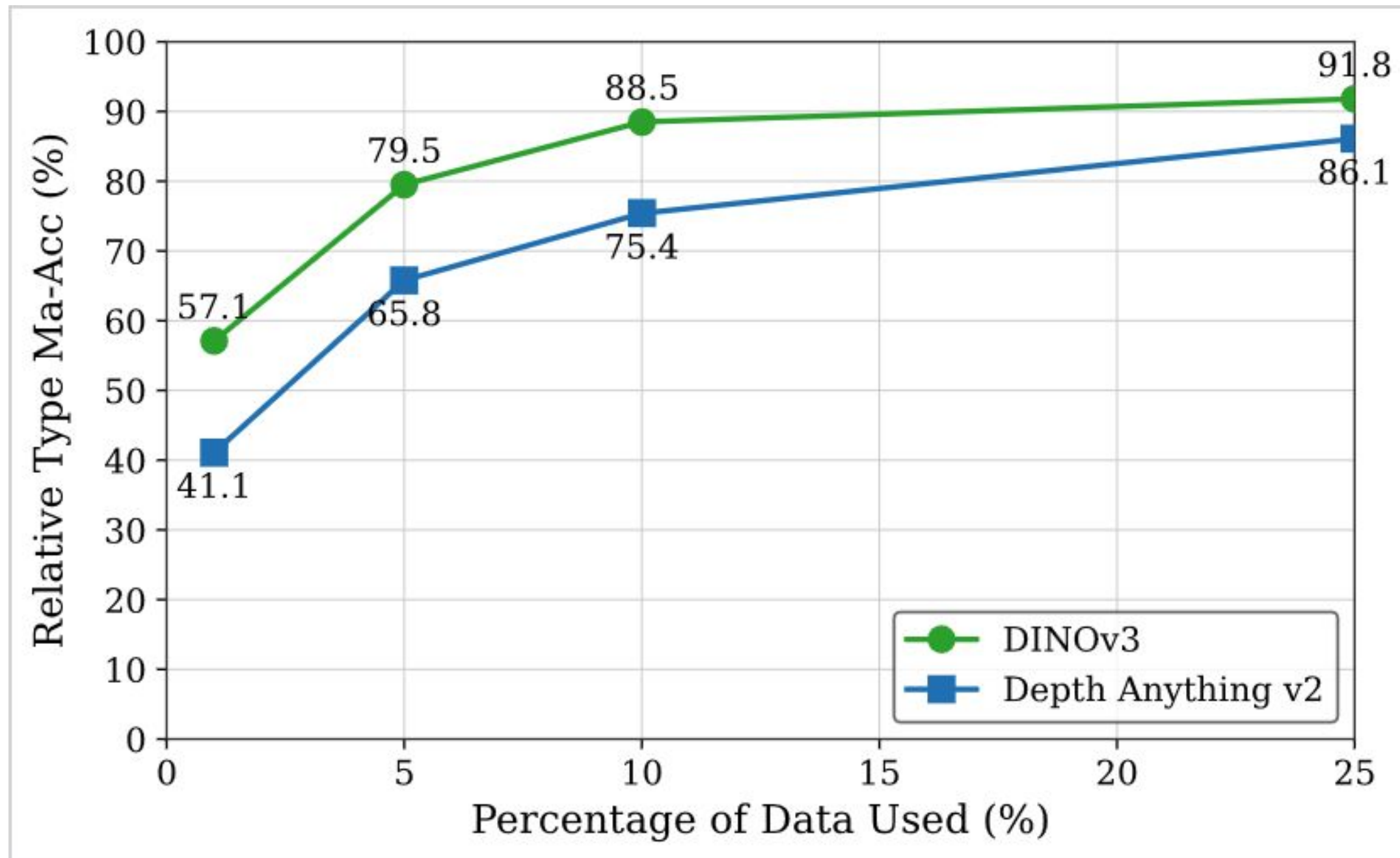


Main Results

Method	Type		Make		Model	
	Ma-Acc	Mi-Acc	Ma-Acc	Mi-Acc	Ma-Acc	Mi-Acc
<i>Best-performing model reported in the original work introducing UFPR-VeSV</i>						
EfficientNet-V2	89.0 (2.0)	96.1 (0.7)	85.0 (1.6)	94.4 (0.6)	86.2 (1.1)	90.9 (0.6)
<i>Best Vision Foundation Model</i>						
DINOv2	95.3 (1.3)	97.1 (0.4)	65.9 (3.8)	73.2 (1.6)	66.5 (1.8)	76.8 (1.3)
DINOv3	97.7 (1.0)	98.7 (0.2)	93.5 (1.1)	97.5 (0.2)	95.0 (0.7)	94.7 (0.6)
<i>Best 3D-Aware Vision Foundation Models</i>						
Depth Anything v1	94.5 (1.4)	96.2 (0.4)	58.2 (3.5)	62.2 (1.9)	61.1 (1.2)	68.8 (1.8)
Depth Anything v2	95.0 (1.7)	96.4 (0.4)	56.6 (4.7)	59.2 (2.6)	64.4 (1.6)	68.1 (2.0)
DUS3R (Encoder)	74.7 (1.9)	79.0 (1.6)	29.1 (3.7)	31.0 (1.6)	25.0 (1.4)	26.4 (2.1)



Robustness: Few-Shot Learning



- Comparison of the best 2D and 3D-aware vision foundation models.
- Linear probes re-fitted using only different fractions of the training data.
- Performance measured on the original test set.

Robustness: Out-of-Distribution (OOD)



Figure: OOD vehicle samples represent vehicle models absent from the original dataset. Images were captured by the same surveillance system used by PMPR.

Model	Ma-Acc (%)	Mi-Acc (%)
DINOv3	89.6 (0.8)	89.7 (0.9)
Depth Anything v2	84.6 (3.3)	88.9 (1.1)

- Ma-Acc: DINOv3 performed better than Depth Anything v2 ($p < 0.01$).
- Mi-Acc: No significant difference ($p \approx 0.084$).

Robustness: Pose & IR sensitivity

Model	Factor	Type	Make	Model
DINOv3	Rear View	1.5	-1.2	7.3
DINOv3	Infrared	-2.6	-8.6	-6.6
DA2	Rear View	0.3	-5.4	3.3
DA2	Infrared	-1.0	-2.0	-2.9

Figure: Conditional accuracy differences (Yes - No) in percentage points.

- DA2 Front vs. Rear: no meaningful accuracy difference ($p = .193$)
- DINOv3 is significantly affected under all evaluated conditions
- Overall, DINOv3 outperforms DA2 in all tasks and conditions.





Conclusions & Future Work

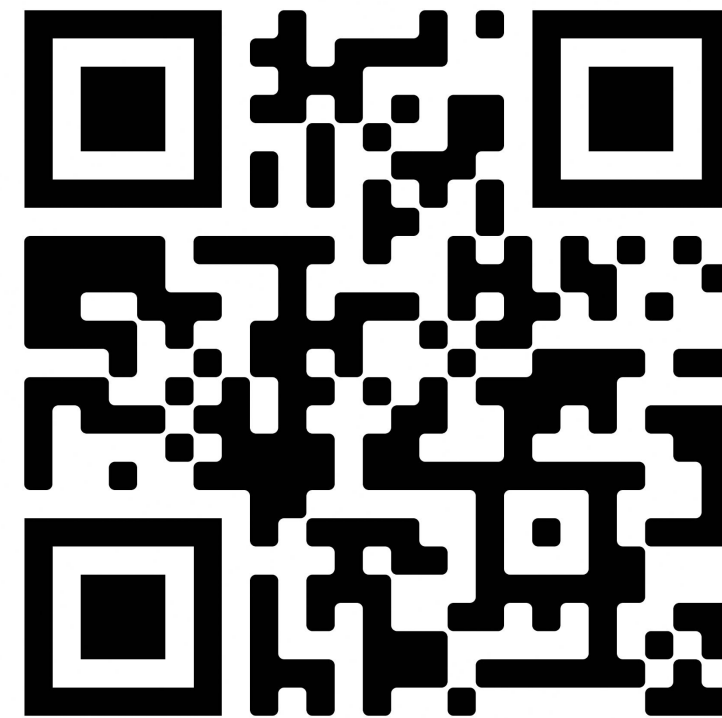
- DINOv3 shows the best performance in all tasks.
- Task granularity matters.
- Depth Anything v2 seems less impacted by pose for Type classification.
- 3D-aware models need further adaptation.
- Current research limitations
 - Confounding factors, such as:
 - Pre-training data scale;
 - Absence of controlled baselines.
- Directions for future work:
 - Better assess performance and data efficiency;
 - Explore hybrid approaches;
 - Expand invariance tests.



Thank You!



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Paper, results, and code: scan the QR code or
visit github.com/UFPR-IPASP-PR/3D-Vision-Benchmark

